

Executive Business Report - July 2026

End-to-End Sales Forecasting & Demand Intelligence System

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Project: End-to-End Sales Forecasting & Demand Intelligence System

1. Executive Summary

This project was developed to help improve sales planning, inventory management, and business decision-making using historical retail sales data. By analyzing four years of sales records, the system identifies sales trends, seasonal demand, unusual sales events, and product demand patterns. It also predicts future sales for the next three months and presents the results through an interactive dashboard. These insights can help management reduce stock shortages, avoid excess inventory, and make better purchasing and supply chain decisions.

2. Key Findings

Sales Performance:

Technology products generated the highest overall sales revenue among all product categories.

Sales showed a clear upward trend over the four-year period, indicating consistent business growth.

Monthly sales exhibited seasonal patterns, with higher demand observed during festive and year-end periods.

Average delivery time remained consistent across most regions, suggesting a stable logistics process.

Forecasting Results:

Three forecasting models were evaluated:

Model	MAE	RMSE	MAPE
SARIMA	18,031	19,009	18.97%
Prophet	20,251	22,318	21.86%
XGBoost	14,443	17,067	14.45%

XGBoost achieved the lowest forecasting errors and was selected as the production model.

3. Three-Month Sales Forecast

The forecasting model predicts that overall sales are expected to remain stable over the next three months, with moderate month-to-month fluctuations that follow historical seasonal patterns.

Rather than indicating a sudden increase or decrease in demand, the forecast suggests continued business stability. Inventory planning should therefore maintain sufficient stock levels while preparing for seasonal demand variations.

The forecasting model also provides confidence ranges, indicating that actual sales may vary due to promotions, market conditions, or unexpected external events.

4. Top Sales Anomalies

Three major anomalies were detected during historical sales analysis:

Anomaly	Possible Business Explanation
Significant sales spike during festive months	Increased customer purchases due to holiday shopping and promotional campaigns.
Sudden decline in weekly sales	Possible inventory shortage or reduced customer demand during off-season periods.
Unexpected short-term sales surge	Promotional discounts, marketing campaigns, or bulk corporate purchases.

These anomalies should be reviewed whenever similar patterns occur in future business operations.

5. Product Demand Segmentation

K-Means clustering grouped products into four demand segments.

Demand Segment	Recommended Stocking Strategy
High Volume, Stable Demand	Maintain higher inventory levels to avoid stock-outs.
Growing Demand	Gradually increase inventory while monitoring future sales.
Low Volume, High Volatility	Keep limited inventory and replenish based on demand.
Declining Demand	Reduce stock levels and avoid over-purchasing.

Demand segmentation enables more efficient inventory allocation and reduces unnecessary storage costs.

6. Business Recommendations

1. Increase Inventory for High-Demand Products

Technology products consistently generated the highest sales revenue. Maintaining higher stock availability for these products can reduce missed sales opportunities.

2. Prepare Inventory Before Seasonal Demand Peaks

Historical data shows recurring seasonal increases in sales. Inventory procurement should begin before these periods to ensure sufficient product availability during peak demand.

3. Use Forecasting for Procurement Planning

The XGBoost forecasting model achieved the lowest prediction error (MAPE: 14.45%). Incorporating these forecasts into procurement planning can improve purchasing decisions and reduce excess inventory.

7. Business Risk / Limitation

This forecasting system is based entirely on historical sales data. Unexpected events such as economic changes, supplier disruptions, policy changes, or large marketing campaigns may significantly affect future sales but cannot be predicted accurately from historical patterns alone.

Regular model retraining with updated sales data is recommended to maintain forecasting accuracy.